

Altitude: The angle between the sun, the observer, and the horizon.

Armillary Sphere

Autumnal Equinox

Azimuth: The compass heading you must face in order to look at the sun.

Celestial Equator: The extension of the Earth's equator to the celestial sphere.

Celestial Pole: The point around which stars rotate.

Celestial Sphere

Declination: The angle between the celestial equator and the star along a line that passes through the star, the north, and the south celestial sphere.

Diurnal Motion

Eccentric Model

Ecliptic

Equator: The largest circumference of the Earth's equator to the celestial sphere.

Gnomon

Horizon

Inclination

Latitude: Circles parallel to the

plane of equator.

Longitude: Circles perpendicular to the equator that pass through the north-south poles.

Meridian: A circle on the celestial sphere that passes through North-Zenith-South.

Nadir

Obliquity

Right Ascension: The angle between the line and the vernal equinox along the celestial sphere.

Sidereal Day

Solar Day

Spring Equinox

Summer Solstice

Tropic of Cancer / Tropic of Capricorn: 23.5 degree North / South. (Remark: Arctic Circle, North Pole)

Tropical Year

Vernal Equinox

Winter Solstice

Zenith

Zenith Passage

Zodiac: The constellations that the ecliptic touches

Apparition: The time that a planet can be observed.

Conjunction: The time when the planets disappear near the sun

Crescent: The moon when less than half is illuminated. (Remark: When half is illuminated, the moon is called quarter moon

Elongation

Eccentric / Epicycle Model: Earth-off-center/Two circles

Evening Star

Far / Near Side

Geocentric Model: With the Earth as a center.

Gibbous

Gregorian Calendar: Modern western calendar

Heliocentric Model: With the sun as a center.

Inferior Planet: Mercury, Venus

Julian Calendar

Julian Year: 365.25 d, but not a leap year!

Lunar/Solar Eclipses

Lunisolar Calendar

Metonic Cycle: 19 years for a cycle when the Solar calendar and the Lunar calendar st

Moon Phases

Morning Star

Near Side

Opposition

Prograde

Retrograde

Seasonal Marker

Sidereal Month: Same position (27.3 d)

Superior Planet: Mars, Jupiter, Saturn

Synodic Month: Complete phases (29.53 d)

Synodic Period: When retrograde/conjunction/opposition happens again.

Tropical Period: When the planet arrives at the same place on the ecliptic.

Waxing: When the moon gets larger

Waning: When the moon gets smaller